

Question

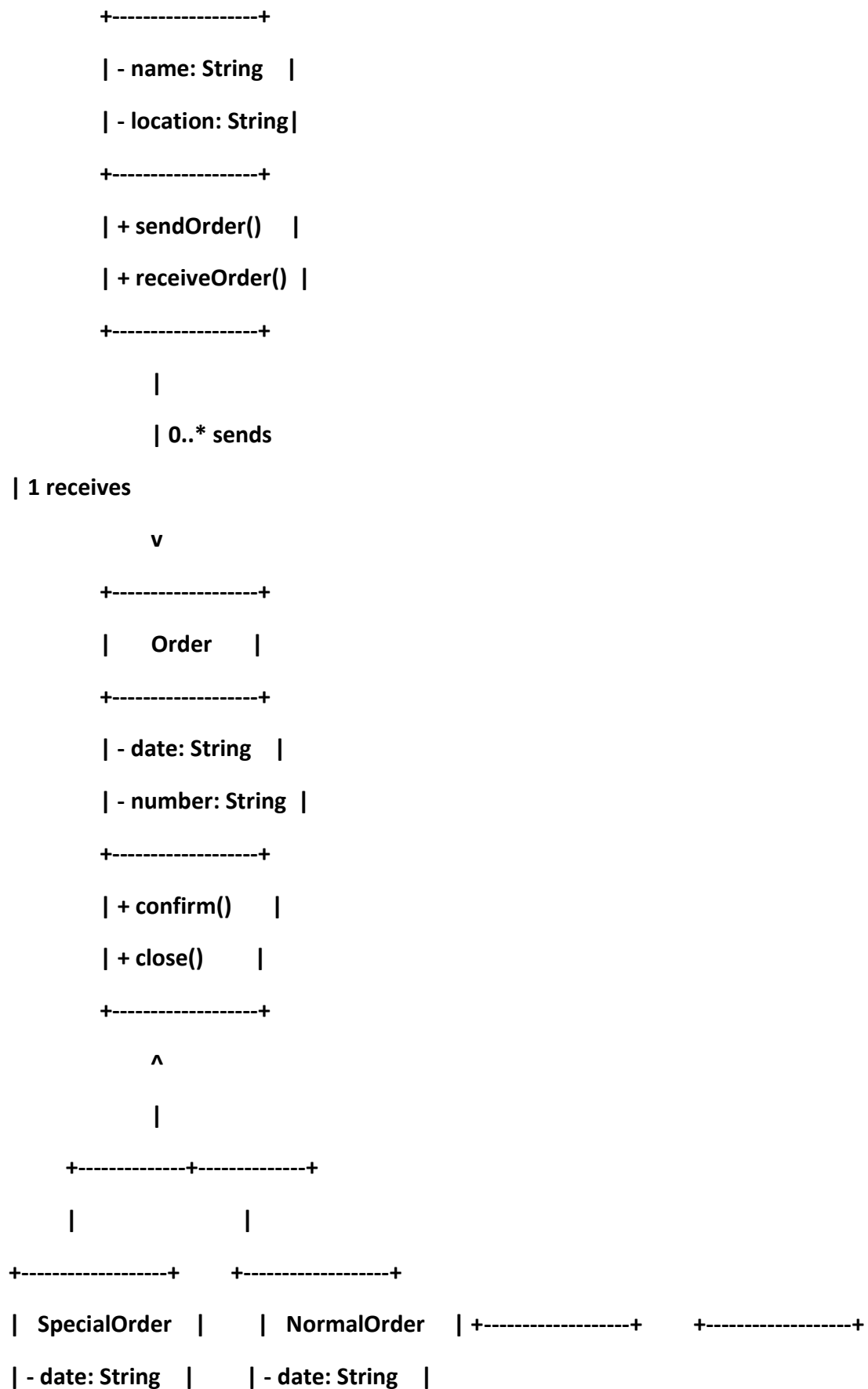
4. Write a Java Program to implement the given class diagram with Customer, Order, SpecialOrder, and NormalOrder classes.

Steps for Execution

1. Open Eclipse IDE (or any Java IDE).
2. Create a new Java project named WeekFourAssignment.
3. Inside src, create a file named weekfouruml.java.
4. Copy the program code into the file.
5. Save the file.
6. Compile and run the program.
7. Observe the output in the console. **Algorithm**
 1. Define a Customer class with attributes name and location, and methods sendOrder() and receiveOrder().
 2. Define a base Order class with attributes date and number, and methods confirm() and close().
 3. Create SpecialOrder class inheriting from Order, adding method dispatch().
 4. Create NormalOrder class inheriting from Order, adding methods dispatch() and receive().
 5. In the weekfouruml class:
 - Create a Customer object.
 - Create SpecialOrder and NormalOrder objects.
 - Call methods to demonstrate sending, confirming, dispatching, receiving, and closing orders.
 6. Display results on the console.

UML DIAGRAM:

```
+-----+
|  Customer  |
```



- number: String	- number: String
+-----+	+-----+
+ confirm()	+ confirm()
+ close()	+ close()
+ dispatch()	+ dispatch()
	+ receive()
+-----+	+-----+

CODE:

```
// Customer class class
```

```
Customer { private
```

```
String name; private
```

```
String location;
```

```
    public Customer(String name, String location) {
this.name = name;    this.location = location;
    }
```

```
    public void sendOrder(Order order) {
        System.out.println(name + " from " + location + " is sending order " +
order.getNumber());
    }
```

```
    public void receiveOrder(Order order) {
        System.out.println(name + " received order " + order.getNumber());
    }
}
```

```
// Base Order class class
Order {    private String
date;    private String
number;

    public Order(String date, String number) {
this.date = date;    this.number = number;
    }

    public void confirm() {
        System.out.println("Order " + number + " confirmed.");
    }

    public void close() {
        System.out.println("Order " + number + " closed.");
    }

    public String getNumber() {
return number;
    }
}

// SpecialOrder inherits from Order class
SpecialOrder extends Order {    public
SpecialOrder(String date, String number) {
super(date, number);
    }
```

```
public void dispatch() {  
    System.out.println("Special order " + getNumber() + " dispatched.");  
}  
}
```

// NormalOrder inherits from Order class

```
NormalOrder extends Order {    public  
NormalOrder(String date, String number) {  
super(date, number);  
}
```

```
public void dispatch() {  
    System.out.println("Normal order " + getNumber() + " dispatched.");  
}
```

```
public void receive() {  
    System.out.println("Normal order " + getNumber() + " received.");  
}  
}
```

// Main class renamed as WeekFouruml

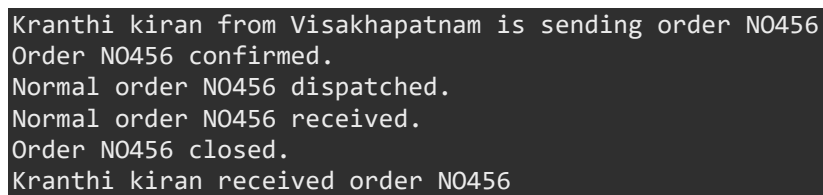
```
public class WeekFouruml {    public  
static void main(String[] args) {  
    Customer c1 = new Customer("Haarika", "Visakhapatnam");    SpecialOrder so =  
new SpecialOrder("2026-08-10", "SO123");  
    NormalOrder no = new NormalOrder("2026-08-10", "NO456");
```

```
        c1.sendOrder(so);  
so.confirm();  
so.dispatch();    so.close();
```

```
        c1.sendOrder(no);  
no.confirm();  
no.dispatch();  
no.receive();    no.close();
```

```
        c1.receiveOrder(no);  
    }  
}
```

Output Screenshot



```
Kranthi kiran from Visakhapatnam is sending order N0456  
Order N0456 confirmed.  
Normal order N0456 dispatched.  
Normal order N0456 received.  
Order N0456 closed.  
Kranthi kiran received order N0456
```